PORTUGAL 

Gravity Model Analysis of Portugal's Bilateral Trade

A Comparison of OLS, PPML and Tobit Estimators

1. Objective

The primary objective of this study is to analyse Portugal's bilateral trade relationships within the framework of the Gravity Model of International Trade. The study examines how economic size, measured by the gross domestic product (GDP) of both Portugal (the exporting country) and its trading partners (the importing countries), together with the geographical distance between them, influences trade flows. Using Portugal's trade data, it seeks to determine whether larger economies tend to trade more and whether distance acts as a barrier to trade through higher transportation and transaction costs.

The analysis further compares three econometric estimation techniques - Ordinary Least Squares (OLS), Poisson Pseudo Maximum Likelihood (PPML) and Tobit regression - to identify which model provides the most accurate and consistent results for trade data that may include zero or limited trade flows.

2. Data Description

The data are drawn from the **BACI International Trade Database** compiled by **CEPII**. BACI is one of the most comprehensive and harmonised sources of bilateral trade data, covering trade flows between more than 200 countries. For this analysis, the dataset was filtered to include Portugal as the reporting (exporting) country, focusing on its trade with multiple destination countries over a given period.

The main purpose of the study is to estimate Portugal's trade flows using OLS, PPML and Tobit techniques. The dependent variable, `tradeflow_comtrade_o`, represents the value of exports or trade flows originating from Portugal, as obtained from the UN Comtrade database. The key independent variables are `gdp_o`, `gdp_d` and `distcap`, denoting the GDP of the origin country (Portugal), the GDP of the destination country, and the geographical distance between their capital cities, respectively. These variables are grounded in the traditional **Gravity Model of Trade**, which predicts that trade between two countries increases with their economic size and decreases with distance.

To improve model specification and interpretability, the variables were transformed into natural logarithms, particularly for the OLS estimation, in order to linearise the relationships and allow the coefficients to be interpreted as elasticities. In the extended OLS, PPML and Tobit models, additional dummy variables - `contig` (common border), `comlang_off` (common official language) and `comcol` (common colony) - were included to capture cultural, linguistic and historical linkages that may also influence trade intensity. All variables were cleaned and processed before estimation, and observations with missing values were dropped to ensure the accuracy and robustness of the results.

Table 1. Variables used in the analysis

Variable	Role	Description
`tradeflow_comtrade_o`	Dependent	Trade flow from Portugal to the destination country
`gdp_o`	Independent	GDP of Portugal (origin country)
`gdp_d`	Independent	GDP of the destination country
`distcap`	Independent	Distance between capital cities (in km)
`contig`	Dummy	Common border (1 = shared border)
`comlang_off`	Dummy	Common official language
`comcol`	Dummy	Common colonial history

3. Model Building and Estimation

3.1 Model 1: Ordinary Least Squares (OLS)

Baseline specification

The first step in estimating Portugal's bilateral trade flows is the **Ordinary Least Squares (OLS)** method. OLS provides a simple and intuitive way to estimate the relationship between Portugal's export flows and the key explanatory variables, economic size (GDP) and distance. The model assumes that trade between two countries increases with their economic size and decreases with geographical distance.

$$\ln(\text{tradeflow_comtrade_o}_{ij}) = \beta_0 + \beta_1 \ln(\text{gdp_o}_i) + \beta_2 \ln(\text{gdp_d}_j) + \beta_3 \ln(\text{distcap}_{ij}) + \epsilon_{ij}$$

Equation 1. Log-linear gravity model

In the context of Portugal, the OLS model captures how export performance varies with Portugal's own economic strength and that of its trading partners, while accounting for the negative influence

of distance. Higher values of GDP_o and GDP_d are expected to be positively associated with trade, as larger economies tend to produce and consume more goods. Conversely, greater values of `distcap` are expected to reduce trade intensity, reflecting higher transportation and transaction costs.

The natural logarithm (ln) of both the dependent and the independent variables is taken to convert the model into a log-linear form. Trade and GDP generally have a multiplicative relationship, and taking logarithms linearises that relationship so that it can be properly estimated by OLS. Moreover, when both sides of the equation are in logarithms, the coefficients can be interpreted directly as elasticities: they show the percentage change in trade flow associated with a one per cent change in GDP or distance, which makes the results easier to interpret in economic terms. The variables were generated in Stata as follows:

```
gen lngdpi = ln(gdp_o)
gen lngdpj = ln(gdp_d)
gen lndistancecap = ln(distcap)
gen lntradeflow = ln(tradeflow_comtrade_o)
```

```
. reg lntradeflow lngdpi lngdpj lndistancecap

      Source |       SS           df       MS      Number of obs   =     9,031
-------------+----------------------------------   F(3, 9027)      =   6543.20
       Model |  64941.131         3  21647.0437   Prob > F        =    0.0000
    Residual |  29864.2572    9,027  3.30832582   R-squared       =    0.6850
-------------+----------------------------------   Adj R-squared   =    0.6849
       Total |  94805.3881    9,030  10.4989356   Root MSE        =    1.8189

------------------------------------------------------------------------------
 lntradeflow |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
       lngdpi |   .3132199   .0162224    19.31   0.000     .2814203    .3450194
       lngdpj |   .9287202    .008485   109.45   0.000     .9120877    .9453528
lndistancecap |  -1.337639   .029098   -45.97   0.000    -1.394677    -1.2806
        _cons |  -1.587412   .3756408    -4.23   0.000    -2.323753   -.8510706
------------------------------------------------------------------------------

.
```

Interpretation

The estimated coefficients are consistent with the theoretical expectations of the gravity model of trade. All variables are statistically significant at the 1% level, and the model explains about **68.5%** of the variation in bilateral trade flows, indicating a very good fit.

The coefficient of **lnGDPi (0.313)** suggests that a 1% increase in Portugal's GDP is associated with an approximately **0.31% increase in its bilateral trade flows**, holding other factors constant. This implies that as Portugal's economic size expands, its capacity to export and import goods also rises, though the elasticity is less than unity.

The coefficient of **lnGDPj (0.929)** indicates that a 1% rise in the trading partner's GDP leads to a **0.93% increase in trade** with Portugal. This nearly one-to-one elasticity implies that Portugal trades substantially more with economically larger partners. The effect of the partner's GDP is

roughly three times stronger than that of Portugal's own GDP, highlighting the dominant influence of foreign market size in determining Portugal's trade volumes.

The coefficient of **lnDistancecap (−1.338)** carries the expected negative sign and is highly significant. It suggests that a **1% increase in the distance between capitals** results in about a **1.34% decline in trade flows**. This magnitude indicates that distance acts as a substantial deterrent to trade, capturing the effects of transportation costs, logistical barriers and weaker business networks between geographically distant partners.

The constant term (−1.587), while statistically significant, has limited standalone interpretation; it represents the baseline level of trade when all explanatory variables take the value of unity in log terms.

Overall result. The results reaffirm the validity of the gravity model in explaining Portugal's trade patterns. Trade increases with the economic size of both partners and decreases with distance, in line with standard international trade theory. The relatively large elasticity of trade with respect to the partner's GDP implies that Portugal's trade is strongly driven by the purchasing power and demand conditions in foreign markets. The high explanatory power ($R^2 = 0.685$) and the statistical significance of the coefficients demonstrate that this simple model captures the key determinants of Portugal's bilateral trade flows effectively.

OLS with log-transformed trade flow and dummy variables

In this stage, the gravity model is estimated by OLS with the dependent variable in logarithmic form. In addition to the core economic variables - the GDP of the exporting and importing countries and the distance between them - the model incorporates a set of dummy variables: contiguity, common colonial history and common official language. These capture qualitative aspects of bilateral relationships that influence trade flows. Contiguity indicates whether two countries share a common border, while common colonial history and common language reflect historical and cultural ties that often facilitate trade by reducing informational and institutional barriers.

By including these variables, the model captures both economic magnitude and non-economic linkages that shape international trade, providing a clearer understanding of how geographical proximity, historical connections and linguistic similarities affect trade intensity.

```
. reg lntradeflow lngdpi lngdpj lndistancecap contig comlang_off comcol
note: comcol omitted because of collinearity
```

Source	SS	df	MS	Number of obs	=	9,001
				F(5, 8995)	=	5527.49
Model	71135.7615	5	14227.1523	Prob > F	=	0.0000
Residual	23152.15	8,995	2.57389105	R-squared	=	0.7545
				Adj R-squared	=	0.7543
Total	94287.9115	9,000	10.4764346	Root MSE	=	1.6043

lntradeflow	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lngdpi	.2616031	.0143855	18.19	0.000	.2334044	.2898019
lngdpj	.9722788	.0075654	128.52	0.000	.9574489	.9871088
lndistancecap	-1.358541	.0266913	-50.90	0.000	-1.410862	-1.30622
contig	-.3022117	.2189428	-1.38	0.168	-.7313894	.1269661
comlang_off	4.490135	.0892679	50.30	0.000	4.31515	4.665121
comcol	0	(omitted)				
_cons	-1.353758	.3379151	-4.01	0.000	-2.016148	-.6913674

Interpretation

The OLS estimation of the log-linear gravity model examines the determinants of bilateral trade flows. The dependent variable, `lntradeflow`, is the natural logarithm of trade flow, while the explanatory variables capture economic size, distance and qualitative bilateral factors.

Both the GDP of the exporting country (**lngdpi**) and the GDP of the importing country (**lngdpj**) have a positive and statistically significant effect on trade flows at the 1% level. A 1% increase in the exporter's GDP is associated with approximately a **0.26% increase** in trade flow, while a 1% increase in the importer's GDP leads to about a **0.97% rise** in trade. This aligns with the theoretical prediction of the gravity model that larger economies trade more owing to higher production and consumption capacities.

The coefficient of **lndistancecap** is negative (−1.3585) and highly significant, implying that geographical distance is a major barrier to trade. A 1% increase in the distance between trading partners reduces trade flows by around **1.36%**, highlighting the role of transportation costs and other frictions associated with distance.

Among the dummy variables, **contig** (common border) is negative but statistically insignificant, suggesting that bordering countries do not necessarily trade more in this sample. In contrast, **comlang_off** (common official language) is positive (4.49) and highly significant, indicating that countries sharing an official language trade substantially more, likely owing to reduced communication barriers and stronger institutional links. The variable **comcol** (common colonial history) is omitted because of collinearity, suggesting that it overlaps with other included variables.

The overall fit of the model is high, with an **R-squared of 0.755**: approximately 75.5% of the variation in trade flows is explained by the included variables. The F-statistic is large and significant (Prob > F = 0.000), confirming that the explanatory variables jointly have a strong effect on trade.

3.2 Model 2: Poisson Pseudo Maximum Likelihood (PPML)

In the PPML estimation, both economic and non-economic determinants of Portugal's bilateral trade are incorporated. The model includes the logarithms of the GDP of the exporting country (Portugal) and of its trading partners, as well as the logarithm of the geographical distance between their capitals, which captures transportation and trade costs. Three dummy variables - `contig`, `comlang_off` and `comcol` - account for the broader socio-historical and cultural dimensions influencing trade. The dependent variable, `tradeflow_comtrade_o`, is the value of exports from Portugal to its partner countries; the independent variables are lnGDPi, lnGDPj and lnDistance, together with the dummies.

In the PPML model, trade flow is used **in levels** rather than in logarithms, because PPML works differently from the log-linear OLS model. Trade data often contain zero trade values between some country pairs, and the logarithm of zero is undefined. Using ln(tradeflow) would force those observations to be dropped, which could bias the results. PPML allows zero trade flows to be retained by keeping the dependent variable in its non-logged form.

A further reason is that PPML assumes the expected value of trade to be an exponential function of the explanatory variables.The model therefore already captures the multiplicative relationship between the variables without taking logarithms of the dependent variable. Logarithmic transformations are applied only to the independent variables, such as GDP and distance, so that their coefficients can still be interpreted as elasticities.

This approach is particularly suitable for Portugal, whose trade data may include several zero values with certain distant or smaller economies. By using the actual trade flow as the dependent variable, those zero-trade cases are not excluded, which makes the estimation more reliable and realistic.

```
. ppml tradeflow_comtrade_o lngdpi lngdpj lndistancecap contig comlang_off comcol

note: checking the existence of the estimates
WARNING: tradeflow_comtrade_o has very large values, consider rescaling
WARNING: lngdpi has very large values, consider rescaling  or recentering
WARNING: lngdpj has very large values, consider rescaling  or recentering

Number of regressors excluded to ensure that the estimates exist: 1
Excluded regressors:  comcol
Number of observations excluded: 0

note: starting ppml estimation
note: tradeflow_comtrade_o has noninteger values

Iteration 1:   deviance =  1.60e+09
Iteration 2:   deviance =  8.41e+08
Iteration 3:   deviance =  6.51e+08
Iteration 4:   deviance =  6.31e+08
Iteration 5:   deviance =  6.31e+08
Iteration 6:   deviance =  6.31e+08
Iteration 7:   deviance =  6.31e+08
Iteration 8:   deviance =  6.31e+08

Number of parameters: 6
Number of observations: 9001
Pseudo log-likelihood: -3.156e+08
R-squared: .87165295
Option strict is: off
```

tradeflow_c~o	Coef.	Robust Std. Err.	z	P>|z|	[95% Conf.	Interval]
lngdpi	.4874137	.0279914	17.41	0.000	.4325516	.5422758
lngdpj	.8367202	.0151075	55.38	0.000	.80711	.8663304
lndistancecap	-1.562791	.0267343	-58.46	0.000	-1.615189	-1.510393
contig	-.2889031	.0665114	-4.34	0.000	-.4192631	-.158543
comlang_off	2.253984	.1860851	12.11	0.000	1.889264	2.618704
_cons	-.6054446	.5000158	-1.21	0.226	-1.585458	.3745683

Interpretation

The PPML results confirm the core predictions of the gravity model of trade. Both the GDP of Portugal (lnGDPi) and the GDP of its trading partners (lnGDPj) are positive and highly significant, indicating that larger economies tend to trade more with each other. A 1% increase in Portugal's GDP is associated with an approximate **0.49% increase** in trade, while a 1% rise in the partner country's GDP increases trade by about **0.84%**. Trade is therefore more strongly influenced by the economic size and demand conditions of partner countries than by Portugal's domestic output.

The coefficient of **lnDistancecap** is −1.56 and statistically significant, in line with theoretical expectations. A 1% increase in the distance between two countries' capitals reduces bilateral trade by about **1.56%**. The negative effect of distance highlights the persistent role of transportation costs, logistical barriers and weaker economic linkages between geographically distant markets.

The contiguity variable (**contig**), which takes the value 1 if two countries share a border, has a negative coefficient (−0.29), which is somewhat unexpected. This may be because Portugal shares a border with only one country, Spain, making the variable almost constant and limiting its explanatory power. In such cases the estimate may not represent a true border effect but rather the idiosyncratic nature of trade with that single partner.

The common official language variable (**comlang_off**) is positive (2.25) and statistically significant, suggesting that sharing an official language can dramatically increase trade flows. The

estimated coefficient implies that countries sharing a common language with Portugal trade nearly **9.5 times more** (exp(2.25) ≈ 9.5) than those without such linguistic ties. This underscores the importance of cultural and linguistic proximity in facilitating communication, contract enforcement and trust between trading partners.

The **comcol** variable was excluded from the estimation owing to collinearity, meaning that its effect could not be separately identified in this model, likely because of its strong overlap with the language and historical-link variables.

The model's explanatory power is high, with a **pseudo R^2 of 0.872**, meaning that the included variables account for nearly 87% of the variation in Portugal's bilateral trade flows.

3.3 Model 3: Tobit Model

The Tobit regression was estimated to account for potential censoring in Portugal's bilateral trade data. The dependent variable, `tradeflow_comtrade_o`, represents Portugal's trade flows with its partner countries. The explanatory variables are the GDP of Portugal (lnGDPi), the GDP of partner countries (lnGDPj), the distance between capital cities (lnDistancecap), contiguity (contig), common colony (comcol) and common official language (comlang_off).

```
. tobit tradeflow_comtrade_o lngdpi lngdpj lndistancecap contig comcol comlang_off, ll(0) vce(robust)
note: comcol omitted because of collinearity

Tobit regression                                Number of obs     =      9,001
                                                F(   5,   8996)   =      80.90
                                                Prob > F          =     0.0000
Log pseudolikelihood = -133947.59               Pseudo R2         =     0.0123

------------------------------------------------------------------------------
             |               Robust
tradeflow_c~o|      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
      lngdpi |   39817.65   4979.876     8.00   0.000     30055.96    49579.34
      lngdpj |   80809.68   4331.671    18.66   0.000     72318.62    89300.74
lndistancecap|   -158950   11138.28   -14.27   0.000    -180783.6   -137116.4
      contig |   4345772   777062.5     5.59   0.000      2822552     5868991
      comcol |          0  (omitted)
  comlang_off|   228137.6   28754.68     7.93   0.000     171771.9    284503.3
       _cons |  -573335.2   110184.6    -5.20   0.000      -789322   -357348.4
-------------+----------------------------------------------------------------
      /sigma |   702551.5   36758.67                      630496.1    774606.9
------------------------------------------------------------------------------
           0  left-censored observations
       9,001     uncensored observations
           0 right-censored observations
```

Interpretation

Both economic-size variables, **lnGDPi** and **lnGDPj**, are positive and highly significant, with coefficients of 39,818 and 80,809 respectively. Larger economies therefore trade more with Portugal, consistent with gravity-model expectations. A one-unit increase in the log of Portugal's GDP raises trade flow by approximately **39,818 units**, while a similar increase in the partner country's GDP raises trade flow by **80,809 units**. The partner country's GDP again shows the stronger impact, reinforcing that foreign market size is a major determinant of Portugal's trade performance.

The coefficient on **lnDistancecap** is negative (−158,950) and statistically significant, confirming that distance reduces trade. Greater physical separation between capitals leads to higher transportation costs and weaker trade linkages, thereby discouraging bilateral exchange.

The **contig** variable has a positive and significant coefficient (434,577), suggesting that Portugal trades substantially more with the country it borders. Since Portugal borders only Spain, this result reflects the strong intensity of trade between the two economies, supported by integrated markets and proximity advantages.

The **common official language** variable (**comlang_off**) also shows a large, positive and significant effect (228,121), indicating that sharing an official language substantially increases trade flows. In the Portuguese context, this reflects strong trade relationships with Lusophone countries such as Brazil, Angola and Mozambique, where linguistic commonality enhances communication, contractual trust and cultural affinity.

Main result of the Tobit model

The Tobit results strongly support the theoretical expectations of the gravity model. Both Portugal's GDP and the GDP of its trading partners have positive and highly significant effects on trade flows. The coefficient for the partner country's GDP is nearly twice that of Portugal's own GDP, suggesting that the size and demand of foreign markets exert a stronger influence on Portugal's trade performance. The coefficient on distance is negative and statistically significant, confirming that greater geographical separation reduces trade, primarily through higher transportation costs and weaker commercial linkages. The contiguity and common-language variables are both positive and significant, pointing to the importance of proximity to Spain and of the trade relationships Portugal maintains with Lusophone countries. The colonial-history variable was omitted because of collinearity, meaning that its influence cannot be separated from that of the other linkage variables.

4. Omission of the comcol Variable in the PPML and Tobit Models

In both the PPML and Tobit estimations, the variable **comcol** (common colony) was automatically omitted by Stata. The Tobit output states that it was dropped "because of collinearity", while the PPML output lists it among the regressors excluded "to ensure that the estimates exist". This consistent omission across both estimations indicates that the problem is not model-specific but arises from the underlying structure of the data.

The `comcol` variable takes the value 1 if two countries share a colonial relationship and 0 otherwise. In gravity models, shared colonial ties often promote trade through institutional, linguistic and cultural linkages. In this dataset, however, `comcol` appears to be perfectly correlated with other dummy variables, particularly `comlang_off` or `contig`. For example, all country pairs with a colonial relationship may also share a common official language, making it impossible for the model to separate the independent effects of the two variables. Stata therefore detects perfect multicollinearity and drops `comcol` so that the coefficients of the remaining variables can be estimated.

Another possible reason is a lack of variation: if most or all observations have `comcol` = 0, the variable contributes no additional explanatory power. In the PPML case, a further possibility is the "separation problem", in which certain combinations of dummy variables perfectly predict zero trade flows; Stata then excludes the variable to avoid infinite or undefined estimates.

The omission of `comcol` therefore does not imply that colonial ties have no influence on trade. It indicates that its effect is already captured by other correlated variables, or that the data do not allow its independent contribution to be statistically identified. The exclusion preserves the stability, validity and interpretability of the estimates for the remaining variables in both models.

5. Conclusion

The OLS, PPML and Tobit regressions collectively highlight consistent relationships between economic size, distance and trade flows. Across all three models, the coefficients of **lngdpi** (log of the exporter's GDP) and **lngdpj** (log of the importer's GDP) are positive and highly significant, indicating that trade between two countries increases with their economic size. This accords with the core intuition of the gravity model: larger economies tend to trade more owing to higher production and consumption capacities. Conversely, the coefficient of **lndistancecap** (log of distance between capitals) is negative and significant in every model, confirming that greater geographical distance deters trade by increasing transportation costs and reducing market accessibility.

The common official language dummy (**comlang_off**) has a strong, positive and significant effect in the OLS, PPML and Tobit models alike, suggesting that a shared language lowers informational and cultural barriers to trade. The contiguity dummy (**contig**) is less stable: it is positive and significant in the Tobit model, negative and significant in the PPML model, and negative but insignificant in the extended OLS model. Because Portugal borders only one country, this variable is close to constant and largely reflects trade with Spain alone, so it should be interpreted with caution. The **comcol** variable was omitted from the PPML and Tobit models (and from the extended OLS) owing to collinearity, most likely because it overlaps with `comlang_off` or `contig`; its effect is therefore already captured by these related variables.

Overall, the three models complement each other by reinforcing the robustness of the core gravity-model determinants - the GDP of the trading partners and distance - while also highlighting the importance of historical and cultural linkages in explaining bilateral trade patterns. The consistency of results across OLS, PPML and Tobit adds credibility to the empirical findings, suggesting that economic size, proximity and shared linguistic characteristics remain the most influential factors driving Portugal's international trade.